Hazem Hassan

Senior AI/ML Engineer — Computer Vision · Production ML
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PROFESSIONAL SUMMARY

Senior Computer Vision Engineer with 9+ years building production, real-time vision systems, and taking algorithms from research to field-deployed product. Hands-on experience with thermal/IR image processing (defogging/dehazing and optical-flow-based anomalous-motion detection), multi-object tracking, and low-latency deployment on embedded NVIDIA hardware. Electrical & Electronics Engineering background, with a track record of optimizing vision pipelines for resource-constrained edge devices (Jetson, TensorRT, ONNX) and leading cross-functional hardware–software integration. Comfortable benchmarking and hardening trackers for dynamic, real-world conditions.

ROLES AND SPECIALTIES

Roles: Senior ML Engineer, AI Technical Lead, Senior Computer Vision Engineer.
Specialties: Computer Vision Pipelines, OCR, Video Analytics, Multi-Object Tracking, Action Recognition, Automated Labeling, Human-in-the-Loop Workflows, Active Learning, Weak Supervision, Dataset Quality Assurance, ML APIs, MLOps, Cloud/Edge Deployment.

CORE SKILLS

Programming: Python (expert); C/C++ (intermediate); CUDA / NVIDIA platforms

Thermal / IR & Image Processing: Thermal/IR image processing, defogging & dehazing, image enhancement & restoration, denoising, optical flow, motion & anomaly detection (familiar with NUC / bad-pixel-correction concepts – ready to ramp on sensor-level calibration)

Object Tracking: Multi-object tracking (MOT), feature-based & optical-flow tracking, deep-learning trackers, ID-switch reduction, tracker tuning for dynamic scenes

Real-Time & Embedded: NVIDIA Jetson, real-time multi-threaded pipelines, low-latency inference, model optimization for constrained hardware, TensorRT, ONNX

ML / Deep Learning: PyTorch, TensorFlow, model training & fine-tuning, custom loss design, evaluation metrics, error & failure analysis

Data Engineering: SQL, PostgreSQL, NoSQL Basics, Pandas, NumPy, Data Cleaning/Transformation, Large-Scale Image/Video/Text Processing, Dataset Preparation Pipelines, Feature Engineering.
Deployment & MLOps: Model Serving, FastAPI, REST APIs, Docker, Kubernetes, CI/CD, Model Optimization, ONNX, TensorRT, Inference Performance Tuning, Monitoring & Failure Analysis.
Cloud & Edge: GCP (Vertex AI), AWS (SageMaker / Ground Truth), NVIDIA Jetson, Real-time Inference, Low-latency Video Pipelines.
Leadership: Technical Leadership, System Design, Cross-functional Collaboration, Mentoring, Code Reviews.

WORK EXPERIENCE

Senior ML Engineer

Beyond AI — March 2024 — Present, Remote, EMEA

- Built an in-house visual-prompt detector (region-prompt contrastive alignment) that adapts to new domains from a few reference images, with no re-training or re-labeling — +3 AP over YOLOE.
- Fine-tuned OpenVLA for task-specific robotic manipulation and quantized the model for edge inference, cutting memory footprint and latency enough to run the policy on-device rather than off-loading to a server.
- Developed thermal/IR image-processing algorithms: fog dehazing/defogging to restore visibility in low-visibility conditions, and optical-flow-based detection of anomalous motion on IR camera feeds.
- Deployed real-time video analytics on NVIDIA Jetson edge devices across 10+ camera feeds / 5+ sites, sustaining 25–35 FPS at <150 ms latency for tracking, counting, and behavior analytics.
- Tuned detection + multi-object tracking for “Easier Gates” flow analytics, reaching 94% flow accuracy and reducing ID switches by 14%.
- Optimized vision pipelines for resource-constrained edge hardware using TensorRT/ONNX, balancing accuracy against latency and compute budgets.
- Built PPE-compliance detection (Aramco) at 94% precision / 90% recall, cutting false alarms 25% via post-processing and threshold calibration.
- Designed active-learning data-selection loops prioritizing high-uncertainty and edge-case samples, cutting annotation effort while improving robustness in dynamic scenes.

Technical Lead

IDefy (Digital Identity/eKYC) — October 2022 — February 2024, Giza, Egypt

- Led digital identity onboarding products (SaaS/SDK/web/mobile), delivering 3 production deployments end-to-end.

- Built validation and quality-control workflows for OCR, face matching, liveness, and identity-verification datasets, improving verification success rate by 20% through data quality tuning and systematic failure analysis.
- Developed configurable OCR/document-processing pipelines that adapted to new structured documents using limited samples, annotation rules, and validation checks, reaching 96% recognition accuracy.
- Analyzed production failure cases across OCR, face verification, and liveness detection to identify noisy data, poor image quality, edge cases, and model retraining opportunities.

Senior ML Engineer (Computer Vision)

RDI — January 2021 — October 2022, Giza, Egypt

- Led the delivery of the Sotoor Arabic OCR System, achieving 12% WER on Arabic manuscripts.
- Built document OCR pipeline modules (layout analysis, segmentation, font detection, recognition, denoising, correction), reducing OCR error by ~7% on noisy scans.
- Adapted the K2 Toolkit for OCR training/decoding, improving training efficiency by ~25% and gaining ~4% WER, boosting sequence recognition stability.

ML Researcher

RDI — January 2017 — January 2021, Giza, Egypt

- Collaborated on a team building NLP systems for the Arabic language across speech and text.
- Developed and deployed production systems, including Speech Recognition, Optical Character Recognition, Text Diacritization, Voice Conversion, and Quran Recitation Verification.

PROJECTS

Agentic Agent Call Center

- Built an agentic AI call-center assistant using LLM tool-calling and Multi-RAG retrieval to automate intent handling, answer generation, and CRM/ticketing actions.
- Designed a reliable conversation orchestration layer with guardrails, escalation-to-human routing, and quality evaluation to ensure safe and consistent customer support automation.

Fashimi APP — Lead Engineer

- Built an AI try-on platform using LLMs for recommendation + prompt generation and diffusion models for avatar generation and photorealistic try-on rendering.
- Developed a recommendation engine leveraging mood/occasion/climate/user preferences to produce personalized outfit prompts.

Valorant Overlay — Developer

- Designed a real-time multi-threaded system extracting gameplay data from Valorant for esports competition overlays.
- Implemented object detection, pattern recognition, and color analysis, achieving a response latency of 50–72 msec.

EDUCATION

Bachelor of Science in Engineering

Cairo University — Electrical Electronics and Communication Engineering · 2016

PUBLICATIONS

Arabic Multi-Genre Broadcast Speech Recognition — Conference Paper, December 2019 — IEEE ASRU 2019
Prognostic AI for Post-Operative Infections — Conference Paper, November 2023 — FCI Annual Conference
QuranScript Verification System — Conference Paper, November 2016 — Sixteenth ESOLE Conference of Language Engineering

HOBBIES

Swimming | Gaming | Padel